

PART -1 : AI CONCEPTS:

Q1: What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science dedicated to building systems capable of performing tasks that typically require human intelligence. Instead of relying solely on rigid, pre-programmed rules, AI systems analyze data, recognize patterns, and make independent decisions or predictions. Ultimately, AI aims to replicate human cognitive functions. This includes capabilities such as learning from experience (adaptation), understanding natural language, solving complex problems, and perceiving environments through visual or auditory inputs.

Q2: Difference between AI, Machine Learning, Generative AI, and Agentic AI ?

While these terms are often used interchangeably, they actually represent nested layers and evolving paradigms within the field of computer science.

Artificial Intelligence (AI): The broadest umbrella term. It encompasses any technique, system, or machine that mimics human intelligence, logic, or behavior.

Machine Learning (ML): A specific subset of AI. Instead of being explicitly programmed with rules, ML systems use mathematical algorithms to learn patterns directly from data and improve their performance automatically over time.

Generative AI (GenAI): A specialized subset of Machine Learning. While traditional ML focuses on analyzing or classifying existing data, Generative AI is trained on massive datasets to create entirely new content, such as text, images, code, or audio.

Agentic AI: An advanced paradigm shift in AI design. Unlike passive GenAI (which only responds when prompted), Agentic AI refers to autonomous systems (agents) that can reason, plan multi-step actions, use external tools, and work independently toward a complex goal with minimal human intervention.

Q3: Explain any three open-source AI models available on Hugging Face

Hugging Face hosts thousands of open-source models. Three highly prominent examples across different modalities include:

1. Llama 3 (Meta)

Type: Large Language Model (LLM)

Purpose: Text Generation and Natural Language Processing (NLP)

Description: Developed by Meta, Llama 3 is a state-of-the-art open-weights model designed for complex text reasoning, coding assistance, translation, and dialogue. It is highly efficient and serves as a foundational base for many custom, fine-tuned AI applications.

2. Stable Diffusion (Stability AI)

Type: Latent Diffusion Model

Purpose: Text-to-Image Generation

Description: This model revolutionised open-source creative AI. It takes text prompts written in plain human language and generates highly detailed, photorealistic, or stylized images. Because it is open-source, developers can run it locally and modify it extensively.

3. Whisper (OpenAI)

Type: Automatic Speech Recognition (ASR) Model

Purpose: Speech-to-Text and Translation

Description: Whisper is trained on a vast and diverse dataset of audio. It can transcribe spoken language into highly accurate text and simultaneously translate multiple languages into English, handling accents, background noise, and technical jargon exceptionally well.

Q4: What is the difference between traditional programming and AI-assisted development?

The main difference lies in how code is generated and who defines the logic:

Traditional Programming: The developer manually writes every line of code, explicitly defining all rules, logic, and syntax from scratch.

AI-Assisted Development: The developer acts as an editor, using natural language prompts to guide AI tools (like GitHub Copilot) that instantly generate, debug, or autocomplete code blocks.

Q5: Explain AI hallucinations with one example

An AI hallucination occurs when an AI model confidently generates an answer that sounds convincing and grammatically correct but is factually incorrect, fabricated, or completely untrue. This happens because AI predicts the most likely next words based on patterns rather than cross-checking actual facts.

Example: Asking an AI to write a biography of a real person, and the AI confidently provides a specific, realistic date and cause of death for someone who is still alive and well.

